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Water Quality Criteria-Stream vs. Effluent Standards
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WATER QUALITY CRITERIA— STREAM VS. EFFLUENT STANDARDS

Harold L. Jacobs, Ira N. Gabrielson, Robert K. Horton, Walter A. Lyon, Earle C. Hubbard, and Gordon E. McCallum

This series of papers composed part of a half-day session dealing with water quality criteria at the Federation's 37th Annual Conference in Bal Harbour, Fla., Sept. 27–Oct. 1, 1964.

Industry Views

Harold L. Jacobs

All uses of water are important to mankind and none are more important than another with one exception. Water for human and animal consumption is, without a doubt, the one which needs consideration above all others.

This paper does not presume to speak for the water needs of all of industry, although it would not be greatly different if it did. These remarks will be limited to the needs of the chemical industry; however, there is no really strict definition of that industry.

Water Supply and Use

The report of the Select Committee on National Water Resources was submitted to the 87th Congress on January 31, 1961, as Senate Report No. 29. This document, generally referred to as the Kerr Report, estimated the average annual stream flow in the United States proper as 1,100 bil gpd (4.2 bil cu m/day). Averages must, of course, take into consideration both highs and lows and there have been years when considerably less water was available. The report estimated that total withdrawals by the year 2000 would amount to 888 bil gpd (3.4 bil cu m/day), or 81 percent of average stream flow. This figure has been publicized extensively without adequate qualification and has given rise to the illusion that the nation soon will run out of water in spite of the fact that the same report denied this. The report pointed out that total consumptive use and depletion, or water disappearance, would amount to 253 bil gpd (0.96 bil cu m/day), or 23 percent of average stream flow.

This consumptive use or water disappearance refers to that water which is used for purposes which prevent it from returning directly or indirectly into the surface streams from which it was taken. It is not lost to the world but most of it must go through

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the cycle of evaporation, cloud formation, and precipitation before it can be used again. Irrigation water is lost to the atmosphere by evaporation and transpiration and to the groundwater table as seepage. Water used for domestic purposes is lost temporarily in lawn sprinkling, clothes drying, and septic tank discharges. A small proportion of industrial cooling water is lost by evaporation whether this occurs in cooling towers or in the receiving water after discharge. This latter loss usually is overlooked but, whenever water cools in a pond or flowing stream, it loses volume by evaporation. Industry also loses some water as a constituent in the products it sells.

Of the 253 bil gpd (0.96 bil cu m/day) listed in the Kerr Report as consumptive use, 223 bil gpd (0.84 bil cu m/day), or 88 percent, were attributed to agriculture and wildlife protection. The remaining 12 percent will be consumed for domestic and industrial purposes with all industrial use including manufacturing, mining, and steam generation accounting for 24.4 bil gpd (0.09 bil cu m/day) or less than 10 percent of the total for all purposes. When placed on the basis of average stream flow, the consumptive use by industry including that for all cooling purposes amounts to a little over two percent. The impact of industry on water resources is, therefore, very small when compared to all other uses. Looking at the water problem in the correct perspective, the specter of water shortage vanishes. The secret of water conservation lies in flow equalization, water reuse for non-consumptive purposes, and quality rehabilitation prior to discharge.

Quality Requirements

While the quality of water is always of significant import, it is often not the most important factor in water use by industry, particularly the chemical industry. There are many cases in which no contact is made between the water and the product. By far the most extensive use is for cooling purposes where available quantity is the prime consideration. In the majority of cases quality matters very little as long as the water is wet. Naturally, industry benefits when solids in suspension are reasonably low and when pH adjustment is not required. When water temperature is too high, however, industry suffers economically because of the increase in required volume. If criteria are to be set up for industrial waters, temperature must be one of them.

The next important use for industrial waters is for process purposes. This category covers a wide range of usage and water quality. Industrial plants tailor their water treatment facilities to the needs of their particular processes which may require only clarification or, in addition, softening. Some processes need water which has been deionized. Water treatment technology has advanced greatly through the years. The development of coagulant aids and ionic exchange materials has simplified greatly the problems. The equipment sizes for specific purposes have been reduced and this saves space and investment.

The water requirements for process purposes are usually a small part of total plant consumption. Two examples which are typical illustrate the general picture in the chemical industry. One of the plants with a cooling-water requirement of 250,000 gpm (946 cu m/min) requires soft water for process purposes of only 3,500 gpm (13.3 cu m/min), or 1.4 percent of the total. Another plant uses 10,000 gpm (37.9 cu m/min) for cooling and has need of 50 gpm (0.2 cu m/min), or 0.5 percent of that as deionized water.

Water for boiler feed requires a special quality. The quantity of such water is usually very small in relation to total plant use. By the use of

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modern technology, boiler feedwater of any required quality can be produced including that superior to distilled water in dissolved-solids content. Here again, the treatment method is tailored to particular power-plant needs depending on boiler pressure, type of boiler, and the extent of makeup required.

In spite of the trend toward automation, people are still necessary for operating industrial facilities. Since they have many of the same needs during working hours that they do at home, they require palatable water of suitable purity for drinking and sanitary purposes including shower water in some instances. For this use, manufacturing plants treat surface waters, pump from wells, or buy from a municipal source in order to save investment. In some plants bottled water is used exclusively for drinking.

Up to this point, no mention has been made of numerical limits of water criteria for industrial purposes. The closest approach to a set of criteria is the one submitted to the USPHS Great Lakes Study Group by a Water Quality Advisory Committee formed at PHS' request. This group prepared a tabulation, "Proposed Guides for Water Quality in the Illinois River Basin," which listed industrial waters in two separate categories: Process and Cooling. The suggested values are as follows:

Characteristic	Process	Cooling
Max. Coliform /100 ml	5,000	—
Min. DO (mg /l)	1.0	—
pH Range	5–9	5–9
Max. Phenolics (mg /l)	1.0	—
Max. Chloride (mg /l)	250	None set
Max. ABS (mg /l)	1.0	—
Max. 5-Day BOD (mg /l)	10	—
Max. Turbidity (mg /l)	250	250
Temperature Desirable (°F)*	<90	<90
Max. NH_3-N (mg /l)	5.0	—
Total Dissolved Solids (mg /l)	500	None set
Max. Color	50	—

* $(°F-32)0.555 = °C$.

While there is no particular fault to find with these values as guides, a comment regarding the chloride content of cooling waters used by the chemical industry is in order. A great deal of stainless steel is used in the industry and it can be damaged severely under certain conditions of temperature and chloride content. The limits which can be tolerated are not defined clearly and there is a variance of opinion concerning acceptable values. This does not mean that high chloride waters are useless for cooling in the chemical industry. It simply means that equipment designers must take account of the cooling-water quality and design for it.

The chemical industry requires and uses water of a quality which covers an extremely wide range from the purest to the dirtiest. In some cases the purity must be of such a high grade that it exceeds that for potable purposes. In others it can use water which is unfit for almost any other use except navigation.

Summary

The chemical industry likes water of high quality just like every other entity in the water use field. The better the water quality the less costly it is to prepare for use, with some minor exceptions. On the other hand, water processing techniques have progressed to the point where a wide

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range of water quality can be adapted to use. The most important need is for quantity.

This discussion is not intended to indicate that the chemical industry considers itself so adaptable that it can ignore water quality. The cost of water treatment is an important item in the balance sheet of any large water user. At the same time, the chemical industry accepts the obligation to discharge water after use in a condition which will not be detrimental to downstream users. This means that, in many cases to suit particular requirements, treatment both before and after use must be carried out and from now on this situation will be with us.

Conservation View

Ira N. Gabrielson

Steps must be taken to ensure that water of adequate quality will be retained in streams and lakes and along the seacoast to support desirable fish and wildlife and to accommodate recreational use. All of us will be losers in many ways if appropriate action is not taken promptly to restore and to retain water quality.

Tide of Pollution

The rising tide of water pollution is an acute and serious matter. The interests of all users of water—not just fish, wildlife, and recreationists—are in jeopardy. Waste treatment capacity continues to fall short of needs despite the investment of $2.3 billion by local, state, and federal governments in community facilities since 1956. More than 80 percent of that huge construction outlay has been made by local and state governments, but about 5,800 communities with a population of nearly 36 million still need new, enlarged, or improved wastewater treatment plants costing $2.2 billion today. Nearly 3,000 communities, housing 5.8 million, need wastewater collection and treatment systems. This is for wastes from municipal sources alone.

Industrial waste treatment needs are not catalogued so neatly. Estimators say that organic wastes from industry will be equivalent by 1970 to the wastes of a population of 210 million. They will contain many toxic materials, in addition. The volume of inorganic wastes is large and unknown as are the concentrations at which they are harmful.

Sanitary and storm sewer lines servicing 40 million people in nearly 2,000 communities also must be separated at an estimated cost of $8 billion. Wastes from agricultural, navigational, and other uses add further to the backlog and to the problem.

Water pollution is a threat to the core of individual, community, and national well-being. Its abatement requires understanding of the levels or concentrations of wastes which are harmful and/or toxic, as well as vigilance, and the continuing investment of time, talent, and money.

If water quality criteria for the various uses are not established and advanced waste treatment methods developed so these criteria can be met, then water pollution will become more acute, because it intensifies with population expansion. It cannot be ignored. It will not flow away.

No individual, no single discipline, has the knowledge or experience to resolve the water pollution imbalance that confronts us today. Its solution requires a team approach and the understanding, the cooperation, and the assistance of everyone.

Some of the principals involved in this large and intricate problem, unfortunately, have not progressed in their thinking beyond the point of

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forcing a community to choose between payroll or pollution. Surely, America is not so bankrupt of ideas and ability that it must choose between its resources. Better economic evaluations are needed. Little thought has been given to the additional costs that downstream communities and industries must pay to remove contaminants from their water source. It is often most economical to remove them at their point of origin.

"A society that allows waste dischargers to neglect the off-site costs of waste disposal will not only devote too few resources to the treatment of waste, but will also produce too much waste in view of the damage it causes," writes Kneese (1).

Recreation Aspects

Rendering large sections of our nation's streams and lakes unfit for desirable fish life and unsuited to many kinds of outdoor recreation is one price society already is paying for ineffective water pollution control.

Water is the lifeblood of outdoor recreation and of the many living resources on which much outdoor recreation is based. Fish live in water and their distribution and number are inescapably influenced by its quality. All waterfowl and other birds and mammals are affected by water quality in one way or another.

The kinds and numbers of fish and other aquatic organisms in streams and lakes are indicators of past environmental conditions and of the quality of water in which they live. Very often, dead and dying fish are the first warning that the water has been impaired. Much more than merely preventing the killing of fish must be done. Determining the factors that inhibit the productiveness of desirable forms of aquatic life, including commercially important shell and fin fish, should have priority in water quality research.

"Public demand for recreation, fish, and wildlife is increasing at a rapid pace, and these resources will be among the first beneficiaries of water pollution control programs, as they are among the first to suffer from pollution," Assistant Surgeon General Gordon E. McCallum told the 11th National Watershed Congress earlier this year.

Water is the central element in many kinds of outdoor recreation in which millions of Americans take part. The President's Outdoor Recreation Resources Review Commission reported in 1962 that "44 percent of the population prefer water-based recreation activities over any others.

"Water also enhances recreation on land," the Commission found. "Choice camping sites and picnic areas are usually those adjacent to or within sight of a lake or stream, and the touch of variety added by a pond or marsh enriches the pleasures of hiking or nature study.

"Recreation on water is increasing," the Commission continued. "This trend is likely to continue as more young people acquire an interest in water sports, new reservoirs are constructed, the boating industry wins new converts, and relatively new forms of water-based recreation, such as skin diving and water skiing, become increasingly popular."

The measure of the outdoor recreation problem—if tending to the recreation desires and demands of the American people can be called a problem—is that outdoor recreation activity, already a predominant influence on American life, is expected to triple by the year 2000.

"This trend will greatly accelerate," the Commission predicted, "if pollution control programs are successful in cleaning up streams, lakes, and seashore areas that are presently off-limits to recreation, or are now so unattractive as to preclude many activities."

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The Commission could have added that the trend will accelerate, too, if ways and means are found to prevent further deterioration of the resource and recreation-reproducing capacities of waters presently uncontaminated or only moderately contaminated by pollutants of many kinds.

A survey by the Izaak Walton League of America disclosed that League members would fish twice as much if pollution was reduced in the waters of densely populated and industrialized states. Several hundred million dollars of potential tourist and recreationist expenditures were not made because of dirty water, the survey inferred.

A principal recommendation of the Outdoor Recreation Resources Review Commission was that recreation should be recognized as a "motivating purpose and as a necessary objective" in the allocation of funds for programs and projects involving water pollution control.

This recommendation has been further expanded by the recent policy declaration of the President's Recreation Advisory Council that "(*a*) recreation be recognized as a full partner with other beneficial water uses in water quality management policies and programs, (*b*) the water resources of the nation be maintained as clean as possible in order to provide maximum recreation opportunities, and (*c*) all users of public waters have a responsibility for keeping these waters clean." The Secretaries of Agriculture; Commerce; Defense; Health, Education, and Welfare; and the Administrator of the Housing and Home Finance Agency are members of the Recreation Advisory Council.

Industry has a sizable stake in the control of water pollution above its personal needs. Industry benefits in myriad ways from the multi-billion dollar expenditures and investments that outdoor recreation creates. Recreation's tremendous contributions to the national economy and to individual health and enjoyment will be greater still in the years ahead, the years expected to yield even higher income and more leisure.

Outdoor recreation and fish and wildlife are confronted with a supply and demand paradox—the demand for water-based recreational opportunity is pyramiding fantastically and the supply of water is limited and unevenly distributed. Pollution eliminates or greatly reduces the recreation-producing capacity of available waters still further.

National water use in 1945 was 160 bil gpd (0.6 bil cu m/day). It increased to 355 bil gpd (1.3 bil cu m/day) in 1963 and is expected to reach nearly 600 bil gpd (2.3 bil cu m/day) by 1980. By the year 2000—the time that outdoor recreation activity is expected to triple—daily use of water could reach 1,000 bil gpd (3.8 cu m/day).

Available water amounts to only 650 bil gpd (2.5 bil cu m/day), and much of it is not located anywhere near the areas of largest anticipated use. Regionally, the reuse of water has become a necessity, and it will become even more widespread in the years ahead.

New Technology Needed

We are on the threshold of need—a threshold of imperative need—for a new water resources technology. There is no alternative. This technology must be aimed at (*a*) helping to abate pollution by preventing contaminants from getting into water supplies, and (*b*) renovating wastewater for reuse.

Man's needs for water tomorrow must not be forfeited by man's indifference toward pollution today. Attention must be centered on improving and protecting the quality of water supplies now available. Industries, communities, and other water users must be spared the necessity of remov-

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ing impurities that should not have reached water supplies in the first place. This involves (*a*) the uniform and accelerated application of present technology, and (*b*) the intensification of research programs of all kinds for the discovery of new methods for improving water quality.

Preventing pollutants from reaching water courses should be adopted as national policy toward water pollution control. Orienting pollution control activities and programs to that objective should be the first order of business.

Failure to enforce pollution laws promptly and uniformly has created a backlog of vested polluters. Some polluters, be they public or private, simply do not care. Others cling to the discredited theory that the discharge of pollutants into streams is a valid and justifiable use of a public resource and, in many places, unfortunately, their views are shared by public health and pollution abatement officials.

Better laws and more vigorous enforcement can help greatly in reducing water pollution, but today's Great Society lacks the technological knowledge to prevent all water pollution. Present methods of municipal and industrial waste treatment remove only a portion of the total polluting load. Only primary treatment facilities are operative in many areas, with secondary plants in some. Even the residues of secondary treatment can impair water quality seriously, especially where they are large in proportion to the receiving stream.

Little is known about agricultural wastes, about the effects of pesticides applied to the land, and about the residues and intermediate products from pesticide processing plants. Waste treatment technology lags behind manufacturing advances. Plastics, detergents, synthetic fibers, and medicines have created new and complex wastes. Some are toxic to fish and aquatic life. Some do not respond to biological treatment and persist in streams for great distances. Others impair the taste of water, stain it, or impart offensive odors.

Effective pollution control depends largely on the acquisition of new knowledge and new techniques that lead to the development of an improved level of water resources management in order to restore, maintain, and improve water quality. The objective is to make it possible, and mandatory, for each water user to return his process or wastewater to the source in a condition suitable for municipal, industrial, agricultural, recreational, and all other uses that may be made of water from that common source.

To reach this stage of compatible water use, the quality of water necessary for each of the uses that is intended must be known. The minimum quality at any point in a water source must be based on the most critical requirements of all the uses to which that water may be put including fish, wildlife, and recreation.

This means that the quality of water that is needed for every water use will have to be determined. Until all these quality requirements are known, it will continue to be difficult, if not virtually impossible, to detect and to designate other than gross or obvious pollution, to measure its undesirability, and to evaluate and recommend control measures.

Water Quality Criteria

Water quality criteria give purpose to pollution prevention efforts. They replace the concept of dirty waters and give reason and meaning to the hitherto undefinable philosophy of keeping waters as clean as possible. Water quality criteria would give municipalities, industries, and all other water users specific goals toward which to work in cleansing waste and process waters that are returned to a common source.

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Once these criteria are known, they will be useful in evaluating different treatment methods for maintaining water quality. They will enable pollution control administrators to keep the public informed and to marshal public support for necessary programs. Industry will know precisely what is expected of it with respect to wastewaters.

Water quality criteria would substitute fact for supposition. Based on adequate research information, criteria can be used by the states as the foundation for more uniform regulations. Broadly accepted and widely applied, the criteria will assist states and communities in refusing the blandishments of water users who seek concessions in waste treatment to the detriment of the overall interest in water resources.

And finally, criteria will assist in the abatement of pollution by providing a firm basis for legal actions. Criteria currently are negotiated with the states in federal enforcement actions.

In supporting water quality criteria as a positive means of controlling water pollution, I want to make clear that I am not advocating stream classification in any sense that would involve the assignment of waste-carrying capacities. Rather, I support the view that present and future known demands for high-quality water for all uses require the monitoring of discharge levels, outfall by outfall if necessary, to see that the quality of the overall water resource for all uses is not destroyed or impaired.

Water quality criteria should not remain fixed and inflexible for all time. They should be upgraded continuously consistent with newly acquired ability to detect wastes, to demonstrate their undesirability at various concentrations, and to cleanse waste and process waters so as to remove impurities. Criteria would be subject to revision as new information and new techniques become available. In application, criteria would be used to work toward the goal of keeping water sources free of contamination for the benefit of all users of water, including fish, wildlife, and recreation.

Achieving the Goal

A high priority should be placed on obtaining the knowledge and establishing the authority for protecting the water resource from further degradation as well as for improving it wherever possible. How this seemingly Utopian situation will be achieved, I do not know. No one knows, in fact, but a number of suggestions have been made. These involve better laws, regulations, and enforcement, new taxes to support more ambitious programs, levies against polluters, and incentive payments to polluters who treat wastes.

Regardless of the procedure or the combination of procedures that ultimately may be used, we must begin to obtain much more information about the efficient treatment of wastes and the water quality criteria for all manner of wastes.

The federal government is starting to work in that direction through its water quality research stations and water pollution control field laboratories. Advanced waste treatment research is getting some support, but appropriations and staffing are grossly inadequate. Much more also must be done by local and state governments and by industry. State programs are poorly financed and about five states appropriate nearly 50 percent of the funds that are invested in all phases of state-level water pollution control each year.

And finally, there must be a significant change in attitude. Perhaps the change already has begun. Karl R. Bendetsen, president of Champion Paper and Fibre Company, had this to say in a speech at the University of Maine Pulp and Paper Foundation meeting in 1964:

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"The handwriting is on the wall in bold script—the American people are deeply perturbed by dirty water; they want clear streams. They will have them.

"We know that the federal government is not for clean water and no jobs. Consequently, there is a sound basis between government and industry for understanding and cooperative effort to find the techno-economic means for both jobs and clean water. This we cannot do alone—we need federal help.

"The federal government is in the area of pollution abatement to stay. The idea that this is primarily a state problem is neither factual, true, nor realistic. We must be prepared to put up very substantial sums of money, and we must convince the federal government that it must do likewise, if our industry is to survive and if we are to have both jobs and clean water."

America's millions of recreationists, fishermen, and hunters support clean water programs. They are willing to cooperate to their utmost with industry and with government at all levels in expanding water pollution control efforts. They know that clean water is the most important criterion for the continuation of their sport.

An Index-Number System for Rating Water Quality

Robert K. Horton

A system for rating water quality in terms of index numbers offers promise as a useful tool in the administration of water pollution abatement. The promise lies in the opportunity to improve techniques for evaluating water quality conditions and for describing water quality in terms of comparative values.

The system proposed is based on the understanding that "water quality" and "pollution" are relative terms, a concept that, although recognized in theory, is not always so readily recognized in practice.

This point is made at the outset because the consideration of water quality and pollution only in terms of absolute measurement is misleading and confusing. Consider, for example, a situation wherein waste control requirements call for maintenance of dissolved oxygen (DO) of, say, 5.0 mg/l in the stream. Interpretation of the requirements in absolute terms leads to the conclusion that with DO less than 5.0 mg/l the stream is polluted and with DO greater than 5.0 mg/l the stream is not polluted. There is no means for expressing gradations in quality other than in terms of "black or white."

Even "stream-classification" systems, which seemingly provide a means for expressing gradations in quality, are actually based on the principle of defining water quality in terms of absolute values. The difficulty with these systems, unfortunately, is that different administrative agencies do not always agree on the same set of values. For example, one agency might establish Class "A" water as water with a coliform count of less than 1,000 MPN (Most Probable Number)/100 ml. Another agency might rule that a count of 2,000 is satisfactory for its Class "A" water. Now, both agencies might agree that water with 1,000 coliforms/100 ml is of better quality than water with 2,000 coliforms/100 ml, but because of the rigidity with which the systems are employed, there is no basis for comparing water quality as designated under one system with that as designated under the other.

In addition, the inflexibility of the system may lead to irrational situations. In the same example, both agencies might designate Class "A"

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water as suitable for recreational purposes. The result would be that water with 2,000 coliforms/100 ml could be used for recreation in one area but not in another.

These considerations lead to this conclusion: First, there is need for a system whereby water quality can be rated on a comparative basis; and second, the system should be designated so as to avoid any connotation of absolute values, such as pollution and nonpollution, or suitability and non-suitability.

A system of rating water quality in terms of index numbers would seem to satisfy this need. It is the purpose of this paper, therefore, to examine the ingredients of such a system and to outline a procedure for determining a water quality index.

A water quality index may be defined as a rating reflecting the composite influence on overall quality of a number of individual quality characteristics.

Development of an Index System

Steps in development of a water quality index are as follows: first, selection of quality characteristics on which the index is to be based; second, establishment of a rating scale for each characteristic; and third, weighting of the several characteristics.

Theoretically, any number of characteristics could go into the makeup of a quality index. However, too large a number would make the index unwieldy. Therefore, it appears practical to use only those characteristics that are of greatest significance. Also, to permit comparison of water quality from one area to another, it seems desirable to select characteristics that are generally significant in most parts of the country. Another limitation is the availability of data; it would be futile to include characteristics that are not generally measured.

In the example of an index system presented here the characteristics selected are those that have been found useful in conducting the activities of the Ohio River Valley Water Sanitation Commission. However, since the characteristics are not unique to streams in the Ohio Valley, they may be regarded as having universal application. At least they may be looked on as primary indicators of quality with the understanding that, if additional refinements are desired, secondary indicators may be added.

Before going further into the selection of quality characteristics, the subject of toxicity calls for special mention. Toxicity is not included in this study on the theory that under no circumstances should streams contain substances that are injurious to humans, animals, or aquatic life. Water containing such substances, therefore, is considered not eligible for index rating.

The quality characteristics selected for examination here, and from which a water quality index will be developed, are shown in Table I.

The first characteristic is designated sewage treatment." This refers to the extent to which municipalities are complying with pollution abatement requirements. The idea is that chemical and biological measures of quality are of little significance until substantial progress has been made in eliminating discharges of raw sewage. No reference is made to the degree of treatment, whether primary, secondary, or other. It is presumed that regulations governing treatment have been established for all sewered communities, and the objective is to measure the amount of sewage load meeting those regulations.

Dissolved oxygen, pH, and coliform count, the next characteristics, are so well recognized as primary indicators of quality as to require no discussion here.

Specific conductance is included because it is a measure of total dissolved solids. The test is an easy one and, perhaps more importantly, it has been

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TABLE I.—Quality Characteristics and Rating Scales for Development of Water Quality Index

Sewage Treatment (% pop. served)	Rating	pH	Rating	Sp. Cond. (μ mhos)	Rating
95–100	100	6–8	100	0–750	100
80–95	80	5–6; 8–9	80	750–1,500	80
70–80	60	4–5; 9–10	40	1,500–2,500	40
60–70	40	<4; >10	0	>2,500	0
50–60	20				
<50	0				

DO (% saturation)	Rating	Coliforms (MPN/100 ml)	Rating	CCE (1×10^{-3} mg/l)	Rating
>70	100	<1,000	100	0–100	100
50–70	80	1,000– 5,000	80	100–200	80
30–50	60	5,000–10,000	60	200–300	60
10–30	30	10,000–20,000	30	300–400	30
<10	0	>20,000	0	>400	0

Alkalinity (mg/l)	Rating	Chloride (mg/l)	Rating	Coefficient
20–100	100	0–100	100	Temperature (M_1) 1 or $\frac{1}{2}$
5–20; 100–200	80	100–175	80	Obvious pollution (M_2) 1 or $\frac{1}{2}$
0–5; >200	40	175–250	40	
Acid	0	>250	0	

Weighting	
Sewage treatment	4
DO	4
pH	4
Coliforms	2
Sp. Cond.	1
CCE	1
Alkalinity	1
Chloride	1

$$QI = \left[\frac{C_1W_1 + C_2W_2 - - - + C_nW_n}{W_1 + W_2 - - - + W_n}\right] M_1M_2$$

automated (as have been tests for several of the other characteristics). Although the correlation is not exact, an examination of available data shows that specific conductance in μ mhos usually runs about 1.5 to 2 times the concentration of total dissolved solids in mg/l.

The next characteristic is carbon chloroform extract (CCE). It is included in order to permit consideration of the influence of organic matter as revealed by this new test. Admittedly, work on the significance of the CCE test may leave something to be desired and there is the further disadvantage that CCE data on streams throughout the country are scanty. However, it seems reasonable to include the CCE test since it is incorporated in the federal drinking-water standards.

The remaining characteristics, as shown in Table I, are alkalinity, chloride, temperature and "obvious pollution." The first three of these are old and familiar indicators and are employed almost universally in evaluation of water quality control.

"Obvious pollution" refers to conditions that are offensive to sight and smell. Such conditions include, among other things, formation of sludge deposits, presence of oil, debris, foam, scum or other floating materials, and discharge of wastes creating a color or odor nuisance. This is the kind of pollution about which the public is

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most knowledgeable. Therefore, an index of water quality should take into account control of the "obvious" if it is to be meaningful from the standpoint of the public.

This makes a list of 10 characteristics that have been selected—for purposes of this example—for construction of an index. The next step is establishment of a rating scale for each characteristic.

Rating Scales

Scales have been chosen so that each characteristic (with exception of temperature and "obvious pollution") can be assigned a value between zero and 100 depending on concentration or quality condition.

For example, with regard to sewage treatment, a value of 100 is assigned if treatment plants are in operation serving virtually all of the upstream population (95 to 100 percent) having direct, measurable influence on the point under consideration. A value of zero is assigned if less than 50 percent of the population is served by treatment plants. Thus, between these limits, rating values may be graded in accordance with the amount of population served. This is shown in Table I.

Assigned ratings for dissolved oxygen, as shown in Table I, are as follows: 100 if DO is greater than 70-percent saturation, zero if DO is less than 10-percent saturation, and with graduated values in between.

All of these rating values are arbitrary and, since they are, there will be differences of opinion as to specific choices. It should be pointed out again, however, that no attempt is being made to measure quality in terms of absolute values. The objective of this system is simply to assign ratings that indicate in a comparative way incremental improvements in quality conditions. The ratings shown are presented for illustrative purposes. In this case they represent the preliminary judgment of the ORSANCO staff. Hopefully, they will offer a basis for discussion through which a consensus might be reached. Suggested rating scales for other characteristics are also shown in Table I.

The last two characteristics, temperature and "obvious pollution," call for a different method of handling. These characteristics, unlike the others, cannot readily be rated to show gradations in quality but fall more into the category of "yes or no" indicators.

Because of the wide variation in the natural temperature of streams, depending on geographic location, it is impossible to assign rating values for different temperature levels, although it is generally acknowledged that the cooler water is the better. There is, however, an upper limit on temperature beyond which deterioration in quality is clearly recognizable.

The ORSANCO Aquatic Life Advisory Committee, a group of expert biologists and fisheries scientists, has recommended that to maintain stream conditions capable of producing a harvestable fish crop of appropriate size and kind, temperature of the stream should not be raised above 93°F (34°C) at any place or at any time.

In view of this recommendation, and because of the detrimental effect of high temperature on other water uses, a temperature of 93°F (34°C) is taken as a cut-off point. The suggestion is that temperature be considered to have no influence on the water quality index for streams having temperatures less than 93°F (34°C), and that an adjustment downward be made in the index for streams having temperatures greater than 93°F (34°C). With regard to the amount of adjustment for temperatures it is suggested that the quality index be taken as one-half of the index determined on the basis of other characteristics.

The procedure proposed for handling "obvious pollution," the last charac-

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TABLE II.—Calculation of Water Quality Index for Two Streams

	Condition	Rating	Weight	Rating times Weight	
Stream 1					
Sewage treatment	96%	100	4	400	
DO	60% sat.	80	4	320	
pH	7.3	100	4	400	
Coliforms	4,000	80	2	160	$QI = \frac{1,660}{18}$
Sp. Cond.	700	100	1	100	
CCE	150	80	1	80	$= 92$
Alkalinity	70	100	1	100	
Chloride	50	100	1	100	
Temperature	<93°F(34°C)				
Obvious pollution	OK				
Total			18	1,660	
Stream 2					
Sewage treatment	75%	60	4	240	
DO	60% sat.	80	4	320	
pH	3.6	0	4	0	
Coliforms	<1,000	100	2	200	$QI = \frac{1,060}{18}$
Sp. Cond.	650	100	1	100	
CCE	90	100	1	100	$= 59$
Alkalinity	0	0	1	0	
Chloride	20	100	1	100	
Temperature	<93°F(34°C)				
Obvious pollution	OK				
Total			18	1,060	

teristic, is the same as that for temperature. If obvious pollution has been eliminated, no adjustment is made; if obvious pollution still exists, the index is taken as one-half of what it would otherwise be.

Weighting and Calculation of Index

The next step in development of a quality index is a weighting of the characteristics to show their relative importance. In the case of temperature and "obvious pollution," this step has already been taken by assignment of coefficients for adjusting the index downward whenever conditions do not meet specified standards.

Of the remaining eight characteristics, upstream sewage treatment, dissolved oxygen, and pH are considered, in the present example, to be of equivalent importance and of more weight than any of the others. Concentration of coliform organisms is considered to be next in significance; following this in order or rank are specific conductance, CCE, alkalinity, and chloride.

To indicate the relative positions of importance and influence on the qualty index, the weighting factors have been assigned on a one-two-four basis. A table of suggested weighting factors is shown in Table I. Again, the assigned weighting factors are arbitrary, and they are presented primarily for illustrative purposes.

This completes the three steps needed for development of an index: selection of characteristics, development of rating scales, and assignment of relative weights. All that remains to be done is to combine the various elements into a single index number. This is done with a simple cumulative formula, as shown in Table I.

As indicated by the formula, the water quality index is calculated by

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(*a*) multiplying the rating points for each characteristic (*C*) by the appropriate weighting factor (*W*); (*b*) adding the weighted rating points and dividing the total by the sum of the weighting factors; and (*c*) multiplying the quotient thus determined by appropriate coefficients (*M*) to adjust for temperature and "obvious pollution" conditions.

The procedure may be illustrated by calculating the quality index for two streams, as shown in Table II. The illustration provides a comparison between conditions in two streams of different quality.

Stream 1 is of very good quality. This is evident from the quality conditions listed in Column 2 of the table: 96 percent of population is served by sewage treatment plants; DO is 60-percent saturation, etc. Column 3 shows rating values for each characteristic in accordance with the scales that have been developed. Note that all characteristics are rated 100 except three: DO, coliform count, and CCE. The next column shows weighting factors, and the last column shows the weighted rating for each characteristic. The sum of the weighted ratings is 1,660, which figure divided by 18 (sum of weighting factors) gives a quality index of 92.

Stream 2 is of poorer quality than Stream 1, mainly because of acid conditions: Alkalinity is zero and the pH is 3.6. Stream 2 is also of poorer quality with respect to the control of sewage discharges: 75 percent of population served compared with 96 percent. Both streams are of about the same quality with respect to DO, specific conductance, and chloride, while Stream 2 is slightly better than Stream 1 with respect to coliform and CCE conditions. The net effect of these varying influences is to produce an overall quality index of 59 which compares with 92 for Stream 1.

Conclusion

The proposed water quality index is intended as a means for comparative evaluation of water quality conditions and pollution abatement programs. Procedures outlined are not considered as a product of final development, but as a starting point from which refinements can be made on the basis of experience. In addition to details that have been presented, others must be examined, such as amount of analytical data required for development of an index and the effect of seasonal variation in stream quality on index ratings.

In the proposed system specific water uses are not necessarily related to any particular range of index numbers. Data at hand indicate that various streams are being put to virtually the same uses although quality characteristics and, hence, index numbers for these streams are widely different. This means, in practical terms, that some areas want and are willing to pay for water with a quality index of 90. Other areas may find that water with an index of 60, for example, is considered satisfactory for their needs.

A quality index system offers a means for measuring pollution abatement progress since stream conditions at any time can be compared with conditions that are desired or planned for the future. In this way the rating system can be useful for administrative purposes and for meaningful communication with the public.

Acknowledgment

Although the author is associated with the Ohio River Valley Water Sanitation Commission, the views set forth in this paper should not be regarded as having any official sanction of the Commission or its advisory committees. Rather, this statement should be regarded as a professional judgment developed by the author in collaboration with Edward J.

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Some Thoughts About Effluent Standards

Walter A. Lyon

In the early days of water pollution control, administrators were confronted with serious pollution problems within their respective jurisdictions and were under great pressure to get the job done in a hurry. In some cases the laws which they were administering did not include requirements concerning the classification of streams and, therefore, the agency was free to decide how it would proceed on the job of deciding how polluted streams were and what should be done to clean them up. While the concept of stream classification is a good and necessary one for water pollution control planning and one that was no doubt used in some fashion in all cases, the degree of detail to which these efforts to classify streams were carried varies greatly and depends in many cases on the specificity of the law in a given area.

In any democratic country regulatory activity is attuned to the concept of equity. This concept is found in many other regulatory endeavors and is one that is always the approach most likely to gain general public acceptance. It is also a concept which very often is difficult to explain or justify in scientific terms. For example, in all states where automobiles are inspected for safety, the same identical inspection is provided for all cars irrespective of their age or condition. From an engineering standpoint, this makes very little sense because it is very likely that an old and poorly maintained car should be given a much more complete and thorough inspection than a new one. Similar concepts prevail in other regulatory activity, such as speed limits, licensing of radio stations, professional licensing, etc.

This concept, too, has found much favor in water pollution control work because in the initial efforts the "basic minimum" approach was used. In Pennsylvania, for example, the engineers of the Division of Sanitary Engineering 30 years ago made general studies of the major streams of the state and, based on this information, set up stream classifications or treatment requirements for entire streams or sectors thereof which required the same degree of removal of BOD and solids from all waste sources irrespective of the volume of the waste. In addition, raw waste standards were established for the major industries which had biologically decomposable discharges. Thus, the concept of equity was also applied to industrial wastes where waste loads and concentrations could vary depending on housekeeping and production.

Some laws and interstate compacts also provided this basic minimum concept. For example, the ORSANCO compact provides that all sewage discharged to waters of the district shall be given a minimum of primary treatment. Similar provisions concerning complete treatment for discharges to clean waters of the Commonwealth are found in Pennsylvania's Clean Streams Law.

Generally, the so-called "effluent standards" established by Pennsylvania and some other states were merely standards based on the size and use of the receiving stream and, therefore, in a sense based on stream standards too. These standards, with some minor exceptions, related to the

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treatment of sewage and biodegradable wastes. They did not relate to toxic wastes, taste- and odor-producing wastes, and other non-biodegradables.

Standards as Technical and Regulatory Criteria

In order to design any type of treatment plant, it is necessary to know what volume and concentration of pollutant may be discharged to the stream. In that sense, it does not really matter whether this effluent standard is based on some standard to be attained in the stream or not.

Also, the enforcement of stream pollution control laws usually makes it necessary to know what may be discharged to the receiving stream. Stream standards are only enforceable when there is no masking of the effect of the effluent on the receiving stream by other upstream discharges of similar pollutants.

Standards as a Guide to Public Policy

The important point which needs to be made here is that the concept of stream standards and effluent standards are not mutually exclusive. In most cases, both are necessary. It is true that the amount of weight being placed on one or the other has, at least in Pennsylvania, depended on the type of situation, the type of waste, and the stage of development of the pollution abatement program. As pollution abatement progress is being made and as the demands for various water uses grow, more and more emphasis will have to be given to the stream standard.

The first phase of work in any pollution abatement program is to take care of obvious pollution problems. The more sophisticated problems can be studied later and, where necessary, additional requirements imposed. It is in this second phase when a detailed study of the stream is necessary that the stream standard becomes of the paramount importance. It is also in this second phase that the question of public policy in the water pollution control effort becomes of the utmost concern. The philosophy of protecting the stream for most uses, even future uses, is a good one from the standpoint of the polluter, the water user, and the public. It permits planned growth of water-using industries and recreation side-by-side, and it prevents undue time lags between the need for water use and its realization.

Now that we have arrived in the age of comprehensive river basin planning, we envision that water quality standards will be based increasingly on this concept that has been familiar to us for a long time—the planned development of a river, not merely the abatement of pollution.

The Dynamics of Water Pollution Control

The dynamics of water quality management are becoming an increasingly important consideration because industrial waste loads and sewage waste loads are changing at an increasingly rapid pace. This is equally true of the hydrology of our rivers because of the growing number of water development projects. The implementation of dynamic water quality management systems will most likely revitalize emphasis on effluent standards since such systems probably will be controlled by water quality and hydrologic forecasts rather than existing stream quality. These standards may differ from the traditional effluent standard because similar wastes from neighboring sources may vary in terms of removal requirements because of differences in volume. The design of such modern water quality systems will maximize social and economic benefits and no longer will be based on equity concepts.

The Role of Standards in Basic Planning

In many situations, the term "pollution" is a relative one, relative to

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the water uses to be protected or enhanced. The effective development of the water resources of a basin makes it necessary that the people who live in the basin decide what their social and economic goals are. These goals and the ability and willingness to pay the price to attain them serve as a baseline for the water resources development of a basin. Water quality standards are an essential part of this plan. They serve as a planning and control mechanism in pollution abatement. They are, in that sense, one place where science and society meet.

It matters not whether stream or effluent standards are used in such a situation. One supplements the other one. Both can be costly or easy to comply with; the two are not mutually exclusive. Stream standards determine the water uses which are protected and enhanced while effluent standards usually are used for purposes of control. Both are tools in the kit of the water pollution control agency and, in the end, tools in the job of protecting the public interest and in attaining the goals, the hopes, and dreams of a modern, complex society.

Stream Standards

Earle C. Hubbard

The relative merits of stream standards vs. effluent standards and other methods of administering water pollution control programs have been subjected to both formal and informal debates for many years and will, undoubtedly, provide a fertile field of discussion for years to come. These debates are, indeed, right and proper inasmuch as the procedures for achieving effective water pollution control are of great importance to all concerned. Certainly, the administrative procedures employed, whatever they are, should be such as to provide a workable basis for applying the fundamental principles of stream sanitation without imposing unreasonable and economically impractical burdens on those who must comply with required pollution abatement measures.

It is not the intent of this paper to review or expand on the arguments for or against the various administrative approaches employed in carrying out water pollution control programs. Rather, it will attempt to present the concept of and the manner in which stream classifications and water quality standards are used in carrying out the water pollution control program in North Carolina. It is believed that this may be done best by reviewing the state's law and program.

Legal Responsibility

In North Carolina the State Stream Sanitation Committee has primary responsibility for the prevention and control of water pollution. The responsibility for administering the program, under the policies of the Committee, has been delegated to the Division of Stream Sanitation and Hydrology of the State Department of Water Resources. The Committee's authority covers the control of pollution in all waters of the state, including both surface and groundwaters which are contained within, flow through, or border on the state or any portion thereof, including those portions of the Atlantic Ocean over which the state has jurisdiction.

State Policy

The State Stream Sanitation Act declares it to be the policy of the state that its water resources shall be utilized prudently in the best interest of the people. It, accordingly, directed the State Stream Sanitation

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Committee, as the official water pollution control agency of the state, to develop a state-wide program designed to protect adequately the present and future water requirements of the state for health, recreation, fishing, agriculture, industry, and animal life.

Concept of Control Program

The act recognizes the philosophy that the state's water resources must be protected for many uses, some of which are in conflict insofar as water quality is concerned. It is, therefore, generally accepted that all streams cannot be maintained in their original state of pristine purity. On the other hand, it is clearly indicated that it is both improper and generally inexcusable to allow streams and other waterways to become degraded below quality standards consistent with the present and foreseeable future uses which they must serve in order to support the water requirements of a given locality or region.

Fortunately, streams have the facility to dilute and assimilate pollution. Therefore, with a reasonable and judicious program of waste treatment, a satisfactory balance between the various stream uses can be obtained. Failure to reach such a balance is wasteful and reflects on the ability of those responsible for water pollution control to understand and apply known waste treatment technologies.

Program Objectives

The fundamental objective of the North Carolina stream pollution control program is that of protecting the receiving stream. It is not removal of BOD, suspended solids, color, or other pollutional materials, per se, but rather the maintenance of stream quality at such levels as to provide appropriately for the present and anticipated downstream beneficial uses. To go beyond this point, simply for the sake of requiring that all wastes be treated to a specified degree, would appear to be impractical from the standpoint of utilizing the stream's assimilative capacity and would result in requiring unnecessary waste treatment facilities.

Approach to the Problem

The North Carolina program is based on and follows the concepts set forth above. The act directed and empowered the Committee:

(*a*) To develop and adopt, after proper study, a series of classifications and water quality standards applicable to each such classification, which would be appropriate for the purpose of classifying each of the waters of the state in such a way as to promote the policy and purposes of the statute most effectively;

(*b*) To survey all of the waters of the state and identify those which, in the opinion of the committee, should be classified separately; and

(*c*) To assign to each identified water of the state such classification, from the series adopted and specified above, as the committee deems proper in order to promote the policy and purposes of the act.

Criteria for Classifications

The act specified that in developing and adopting classifications, and the standards applicable to each, the Committee should recognize that a number of different classifications should be provided (with different standards applicable to each) so as to give effect to the need for balancing conflicting consideration as to usage and other variable factors; that different classifications with different standards applicable thereto may frequently be appropriate for different segments of the same water; and that each classification should be adopted with primary reference to an existing or a contemplated best usage to be made of the waters in question.

The first step taken in formulating the program, therefore, was the devel-

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opment of a system of classifications and water quality standards. This was accomplished through the use of an unofficial advisory committee, comprised of representatives of the state's universities and other state agencies, together with persons representing the agricultural, industrial, and municipal interests of the state. The classifications and standards recommended by this group were then presented to the public during five public hearings held across the state, at which time the philosophy of the program was also explained. The comments of the public at these hearings were recorded and later served the committee well in finalizing the system of classifications and water quality standards adopted for use in classifying the waters of the state. These classifications and a summary of each are as follows:

Fresh-water Classifications	Best Usage
A-I	Unfiltered public water supply after approved disinfection.
A-II	Public water supply with approved treatment equal to coagulation, sedimentation, filtration, and disinfection.
B	Bathing, skiing, and similar aquatic recreation.
C	Fish and wildlife propagation.
D	Agriculture and industrial cooling and process water, after such treatment as may be required.
E	Navigation and waste disposal.

Salt Water Classifications	Best Usage
SA	Shellfish propagation for market purposes.
SB	Bathing, skiing, and similar aquatic recreation.
SC	Salt water fish propagation.
SD	Navigation and waste disposal.

Criteria for Standards

In establishing the standards applicable to each classification, the Committee was required to consider the extent to which any physical, chemical, or biological properties should be prescribed as essential to the contemplated best usage. This was done in formulating the pertinent water quality standards which were assigned automatically upon the classification of waters according to the determined present or future best usage. The standards were designed specifically to provide water of suitable quality for the best usage indicated by the assigned classification, as well as for any other usage requiring water of a lower quality. The term "pollution" is, therefore, defined to mean a condition of any waters which is in contravention of the standards established and applied to such waters in accordance with the classification system.

Assignment of Classifications

The mere act of adopting a series of classifications, with their accompanying water quality standards, did not actually classify any waters of the state. This was done later after complete and detailed studies were made in each of the state's sixteen major river basins to determine and catalogue (*a*) the physical characteristics of the streams, (*b*) the character of the district bordering the streams, (*c*) the uses which have been or may be made of the waters, (*d*) the extent to which such waters were already receiving wastes and the economic values involved in improving their conditions, and (*e*) suggested classifications which, in the opinion of the Committee, would be appropriate for assignment to the waters involved. Public hearings were held in each major river basin for the purpose of presenting information gathered during the studies and to obtain public comment relative to the suggested classifications for the waters of the basin. The last of the 16 river basins was classified in 1963, thus completing the comprehensive study and classification program for the entire state.

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Comprehensive Pollution Abatement Plans

Following the assignment of classifications to the waters of each major river basin and the establishment of the "effective date" for application of the enforcement sections of the statute, a comprehensive pollution abatement plan was issued. This plan attempted to evaluate each significant source of pollution with respect to its volume, strength, and adequacy of any existing treatment facilities employed. The effect of such waste discharges on downstream water uses also was evaluated and, in general terms, the action required of the person or persons responsible for waste discharges was defined. Perhaps of even greater importance, the plan also sought to give meaning to the concept that pollution abatement, like other river basin development, should be comprehensive in scope if it is to result in maximum public benefits.

Voluntary Action Encouraged

It is the policy of the State Stream Sanitation Committee to promote and encourage voluntary action in the abatement of pollution. To effect this policy, each polluter was given the opportunity of submitting, within six months after receipt of the comprehensive pollution abatement plan, an acceptable proposal setting forth (*a*) the intent to comply voluntarily, (*b*) steps which will be taken in relation to providing required pollution abatement measures, and (*c*) a schedule for taking the various steps necessary for the accomplishment of adequate waste treatment or control facilities. If such proposal was acceptable to the technical staff, it was recommended to the Committee, at which time it issued a temporary permit allowing the polluter to continue discharging wastes while pollution abatement measures were being taken. New waste discharges may not be established in a classified river basin until and unless the owner has applied for and obtained a permit from the Committee. In the event progress is not made toward pollution abatement at existing points of waste discharge, the Committee is authorized to revoke any temporary permit held and issue, after public hearings, orders requiring necessary pollution abatement action. Failure to comply with orders of the Committee subjects the holder to court action. In all cases in which such orders have been issued to date desired results have been achieved without the necessity of court action. It is earnestly hoped that this record may be maintained; however, if the necessity arises, the Committee is empowered to and will institute enforcement action in the proper court as required by the statute.

Need for Flexibility

As indicated earlier, the primary objective of regulatory activity in pollution control, including construction of waste treatment facilities, is the protection of the receiving stream for one or more specified beneficial uses. Inasmuch as assimilative capacities of streams change with variations in flow, temperature, and other characteristics and, since the beneficial uses also vary with changing water-use requirements, the program must include flexibility to accommodate these variations. Failure to provide for changing conditions would, and often does, constitute a serious pitfall for the unwary regulatory agency.

The North Carolina statute recognizes this necessity and authorizes the Committee to reclassify any of the waters of the state commensurate with changes in use requirements. It also authorizes the issuance of permits with such conditions and limitations as the Committee deems proper to achieve the purposes and objectives of the program and to modify or revoke such permits upon failure of the holder to comply with any condi-

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tions and limitations attached thereto. Permits are generally issued for specified periods of time, but in no cases are they issued for periods in excess of 10 years.

In general, waters are reclassified upon the request of an existing or prospective user; however, the committee may do so on its own motion. In either case, public hearings must be held prior to changing the classification of any waters or the water quality standards applicable to any classification. This procedure insures the public its rightful opportunity to comment on the proposed changes and, in actual practice, makes the public a party to whatever action is taken by the Committee.

Progress in Pollution Abatement

Progress in the state's water pollution control program to date has been admirable and demonstrates the willingness of those responsible for waste discharges to cooperate in an effort to preserve our water resources. Municipalities and industries have, for the most part, moved forward with the construction of waste treatment and control facilities on a voluntary basis. Those which have not made satisfactory progress in keeping with the terms and conditions of temporary permits issued by the Committee have been issued special orders or have entered into consent judgments which set forth the conditions for compliance.

Since initiation of the program in 1951, the municipalities and industries of the state have either completed or have under construction 703 waste collection and treatment projects costing $142 million, while final plans for 10 additional projects, estimated to cost $4.7 million, have been approved for early construction. In this connection, it is of interest to note that the rate of annual expenditures for waste collection and treatment works has increased from approximately $5 million in 1952 to $22.8 million in 1963. This represents an increase of more than 400 percent in terms of money being spent annually for pollution control facilities.

Conclusion

There are several different approaches employed by regulatory agencies in the administration of water pollution control programs. Each has its own peculiar advantages and disadvantages and, under appropriate circumstances, all may be considered as having a legitimate place in the pollution control field. In the final analysis, however, the ultimate objectives of the administrative approach, or combination of approaches, should be the protection of the receiving waters for an existing or contemplated beneficial usage.

The system of stream classifications and water quality standards has proven to be a valuable tool in the development and execution of the North Carolina program. It provides for flexibility in the adoption of methods for pollution control and abatement while, at the same time, placing the primary emphases where they should be, i.e., on the receiving stream. It must be recognized, however, that the use of stream standards requires a thorough knowledge of stream conditions and uses, which knowledge can be obtained only through exhaustive and time-consuming studies. This dictates a strong, well staffed, and adequately financed pollution control agency which is so closely associated with local problems as to permit adjustments in the classifications and standards in keeping with changing stream conditions and water requirements.

The adoption of stream standards as a tool for regulatory control of pollution is not without limitations. For instance, the application of rigid standards to large areas may well become a barrier to orderly economic development, thus defeating some of the benefits to be derived from the equitable use of the water resources within a given locality or region. Likewise, the establishment of water

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quality standards which are too high, or the continuance of such standards in effect, may well place an unnecessary burden upon those responsible for pollution abatement. For these reasons, the use of stream standards should be attempted only at levels of government which are alert to local needs and which can assure the flexibility in administration necessary to the most equitable usage in the public interest.

Water Quality Criteria Development

Gordon E. McCallum

The people have every right to know what is happening to the nation's waters. They have a right to know who is polluting the streams, with what, what steps are being taken to curb this pollution, and how fast we are reaching the goal of best use of water resources.

The Public Health Service is required to inform the Congress from time to time of the progress being made in combating water pollution. We report on the number of municipal waste treatment plants built with the aid of federal funds, the progress being made on the major river basin programs, and the results of research work and investigations.

But these progress reports do not tell the whole story. Progress in water pollution abatement must really measure the actual end-product, the condition of the nation's streams, lakes, and estuaries. This cannot be done properly until there are better criteria of water quality and more adequate knowledge of what water should be.

The lack of knowledge on which to base criteria works a handicap on the users of water who have a right to know what their responsibilities are. The National Conference on Water Pollution in late 1960 expressed the conviction that "the goal of pollution abatement is to protect and enhance the capacity of the water resource to serve the widest possible range of human needs, and that this goal can be approached only by accepting the positive policy of keeping waters as clean as possible. . . ."

The public generally subscribes to this doctrine, but difficulties come as soon as someone tries to define the phrase "as clean as possible." How should the state or other regulatory body enforce the rule that the waters shall be kept "as clean as possible?" How should the water user repair the damage he does to the water?

Almost everybody agrees that "ultimately we should come to the characteristics of the receiving water as our measure of pollution or our measure of progress." Unfortunately, however, this will not be easy.

Drinking Water Standards

It has not been easy to reach even the present stage. The PHS has been setting drinking water standards since 1914. Though difficult, this is a far easier job than setting standards for the quality of in-stream waters because the only water use being considered is that of public water supplies. There are only two objectives in view: the protection of health and the potability of the water. With in-stream water there are a hundred uses, many of them competing and, consequently, a wide range of objectives also may be competing.

Standards for drinking water have changed over the years under the influence of new scientific knowledge and new environmental hazards. These changes will continue in the future, affecting not only the existing standards for drinking water, but the evolving standards for in-stream waters.

When the 1925 edition of the standards was written, fluoride concentrations of several mg/l were going un-

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detected. Subsequently, it was learned that dental fluorosis was caused by these high concentrations and, when the 1942 standards were issued, the fluoride limit was lowered. By 1946, water treatment could provide optimum fluoride concentrations which would not only prevent dental fluorosis but provide benefits by reducing dental cavities. The 1962 standards recognized this new information and included a detailed set of standards for fluoride concentrations.

Owing to the increasing number and complexity of new types of pollutants in our waterways, drinking water standards have undergone other changes. In 1962, for instance, a limit was set for ABS, the active ingredient in most synthetic detergents. In 1962 a section on radioactive materials was added for the first time.

Future Criteria

The splendid work (2) by McKee and Wolfe points the way for future development of revised in-stream standards. This publication organizes and presents present knowledge on the effects of pollutants on water uses as found in the literature, regulations of state and interstate agencies with regard to the water quality standards they use, what the courts have said about water quality standards and how they have used them in deciding water pollution cases, and the usual practice in evaluating suitability of water for various uses.

This volume, in its earlier and present editions, has been extremely helpful in showing the development of water quality criteria.

An increasing number of parameters is being added to the surveys conducted by the Public Health Service's Water Pollution Surveillance System, formerly called the National Water Quality Network. It, too, is grappling now with the task of identifying the many synthetic substances, pesticides, radioactive elements, and complex chemicals that are appearing in the water resources. It appears, however, that the modern chemical industry continues to develop other compounds faster than methods for identifying them and evaluating their effects can be developed.

New methods of measuring many pesticides are leading to some answers unknown even a year ago. A recent dramatic accomplishment consisted of finding the cause of massive fish kills in the lower Mississippi River. Researchers are using gas chromatography among new techniques that enable them to measure trace amounts of matter in parts per trillion instead of mg/l. As a result, "post-mortems" may be carried out on fish to identify the substances which killed them. These steps in determining the sources and effects of pollution must usually precede measures of control.

Following are a few accomplishments which indicate progress in the development of criteria:

1. A catalog of information on the toxicity for fish of various substances, including metal salts, metallo-cyanide complexes, detergents, pesticides, industrial organic chemicals, and selected industrial waste effluents;
2. Various methods of determining the effects of sublethal concentration of certain toxicant, heavy metals, cyanides, and ABS on the life-history stages of fishes, aquatic insects, and bottom-dwelling organisms;
3. A standardized bioassay procedure for measuring the toxicity of specific chemicals and effluents to fish;
4. Toxic levels to fish, insects, crayfish, and certain benthic organisms of the 10 most widely used chlorinated hydrocarbon pesticides, 15 most common organic phosphorous compounds, 100 widely used herbicides and algicides, 18 principal organic solvents, the principal nitrite compounds, and other materials;
5. Basic knowledge on the environmental requirements for the propagation of fish from studies to determine DO requirements as related to temperature, dissolved gases, pH, and

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other environmental factors such as stream current and diurnal fluctuations of light. These and other studies will lead to the establishment of water quality requirements for fish and other aquatic life; and

6. Basic knowledge on the role of algae in producing tastes and odors in water, and knowledge of the role of nutrients in algal growth.

Research in all phases of water quality criteria will receive special attention when two special laboratories devoted to this subject are opened by the PHS. The freshwater laboratory will be located at Duluth, Minnesota, while the salt water laboratory will be at Kingston, Rhode Island. Each will be staffed with 50 to 75 scientific and professional personnel.

The work in these laboratories will be carried on in close cooperation with university scientists and engineers expert in the water quality field, and with representatives from industry, agriculture, conservation organizations, and state and local governments.

Outside research programs will be encouraged and supported by grants and contracts for research and demonstrations. The work of the Duluth laboratory should be stimulated by its proximity to the Duluth campus of the University of Minnesota, where extensive studies are being conducted on the biology of streams and lakes; to the Oliver Research Laboratory of the U. S. Steel Corporation, also in Duluth; and to Wisconsin State College, in nearby Superior. Similarly, the Kingston marine waters laboratory will work closely with the Narragansett Marine Laboratory of the University of Rhode Island and with the laboratory of the Fish and Wildlife Service, Department of Interior.

They will work on selected aspects of water quality criteria needed to define the adverse effects of pollutants on water uses. They will produce information needed by water pollution control agencies for setting standards which will describe water most suitable for these uses.

Generally speaking, the water quality laboratories will develop reliable methodology to detect and identify water pollutants and measure their influence on the quality characteristics which determine the suitability of water for different uses. The laboratories will make chemical and bioassay studies of wastes to determine their fate and toxicity in the freshwater and salt water environments, particularly as they affect fish, shellfish, and other aquatic organisms, and establish the tolerance levels of fish to specific pollutants. They will receive, evaluate, and continuously review water quality standards previously accepted by government bodies and other agencies, comparing the latest information on the effects of pollutants with earlier findings, and recommending the changes needed. Finally, the water quality laboratories will store and retrieve pertinent research findings in water quality obtained from sources throughout the world.

Conclusion

So, although setting standards for in-stream waters will be a difficult and painstaking job, a start is being made. With the information that has been gathered patiently by the Water Pollution Surveillance System since 1957, with the broad research program conducted by the Public Health Service and educational institutions, and with the promise of new emphasis from two new national water quality laboratories, the task of seeking the information necessary to establish criteria is progressing.

References

1. Kneese, A. V., "The Economics of Regional Water Quality Management." The John Hopkins University Press, Baltimore, Md. (1964).
2. McKee, J. E., and Wolfe, H., "Water Quality Criteria." California State Water Pollution Control Board, Sacramento (1963).

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